### 5.2.1 Collinearity of three points

If three distinct points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are collinear, then we cannot form a triangle, because for such a triangle there will be no altitude (height). Therefore, three points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ will be collinear if the area of $\Delta ABC = 0$.

**Note**

**Another condition for collinearity**

If $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are collinear points, then

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

or $x_1y_2 + x_2y_3 + x_3y_1 = x_1y_3 + x_2y_1 + x_3y_2$.

Similarly, if the area of $\Delta ABC$ is zero, then the three points lie on the same straight line. Thus, three distinct points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ will be collinear if and only if area of $\Delta ABC = 0$.

## 5.3 Area of a Quadrilateral

If $ABCD$ is a quadrilateral, then considering the diagonal $AC$, we can split the quadrilateral $ABCD$ into two triangles $ABC$ and $ACD$.

Using area of triangle formula given its vertices, we can calculate the areas of triangles $ABC$ and $ACD$.

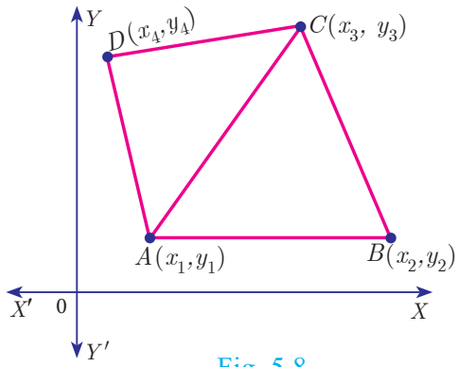



Fig. 5.8

Now, Area of the quadrilateral $ABCD$
= Area of triangle $ABC$ + Area of triangle $ACD$

We use this information to find area of a quadrilateral when its vertices are given.

**Thinking Corner**

How many triangles exist, whose area is zero?

Let $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ and $D(x_4, y_4)$ be the vertices of a quadrilateral $ABCD$.

Now, Area of quadrilateral $ABCD$
= Area of the $\Delta ABD$ + Area of the $\Delta BCD$ (Fig 5.9)

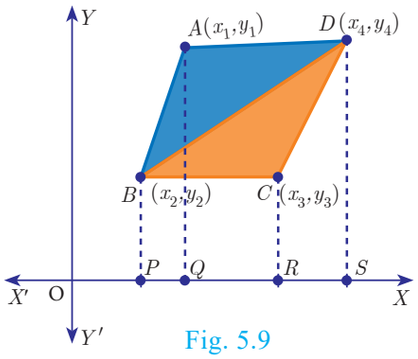



Fig. 5.9

$$= \frac{1}{2}\left\{(x_1y_2 + x_2y_4 + x_4y_1) - (x_2y_1 + x_4y_2 + x_1y_4)\right\}$$

$$+ \frac{1}{2}\left\{(x_2y_3 + x_3y_4 + x_4y_2) - (x_3y_2 + x_4y_3 + x_2y_4)\right\}$$

$$= \frac{1}{2}\left\{(x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (x_2y_1 + x_3y_2 + x_4y_3 + x_1y_4)\right\}$$

$$= \frac{1}{2}\left\{(x_1 - x_3)(y_2 - y_4) - (x_2 - x_4)(y_1 - y_3)\right\} \text{ sq.units.}$$

The following pictorial representation helps us to write the above formula very easily. Take the vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ and $D(x_4, y_4)$ in counter-clockwise direction and write them column-wise as that of the area of a triangle.



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